DIC_Assignment 3

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1 Problem A

For the following Karnaugh map, give the circuit implementation using one 4-to-1 multiplexer and as many 2-to-1 multiplexers as required, but using as few as possible. You are not allowed to use any other logic gate and you must use a and b as the multiplexer selector inputs, as shown on the 4-to-1 multiplexer below.
Implementing just the portion labelled top_module, such that the entire circuit (including the 4-to-1 mux) implements the K-map.

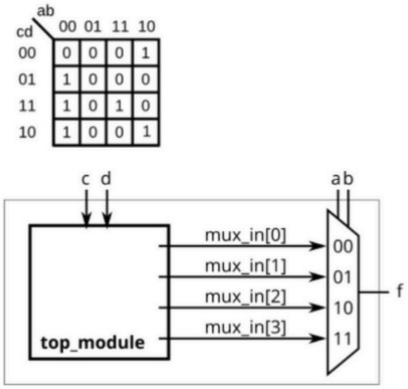

Figure 1: Problem A

1.1 Steps

- Design 2-to-1 Mux instance
- Design top_module

1.2 Implementation

- 2-to-1 MUX

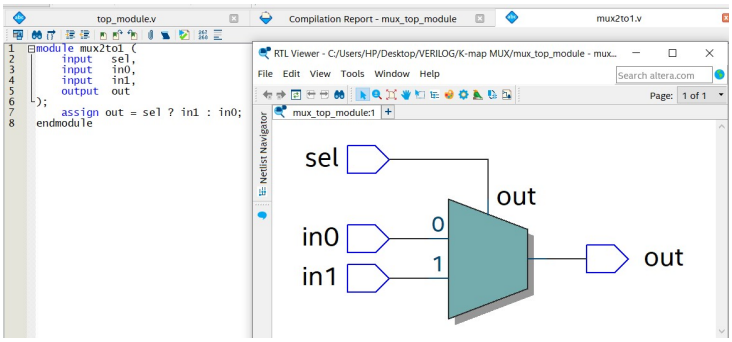

Figure 2: 2-to-1 MUX

- top_module

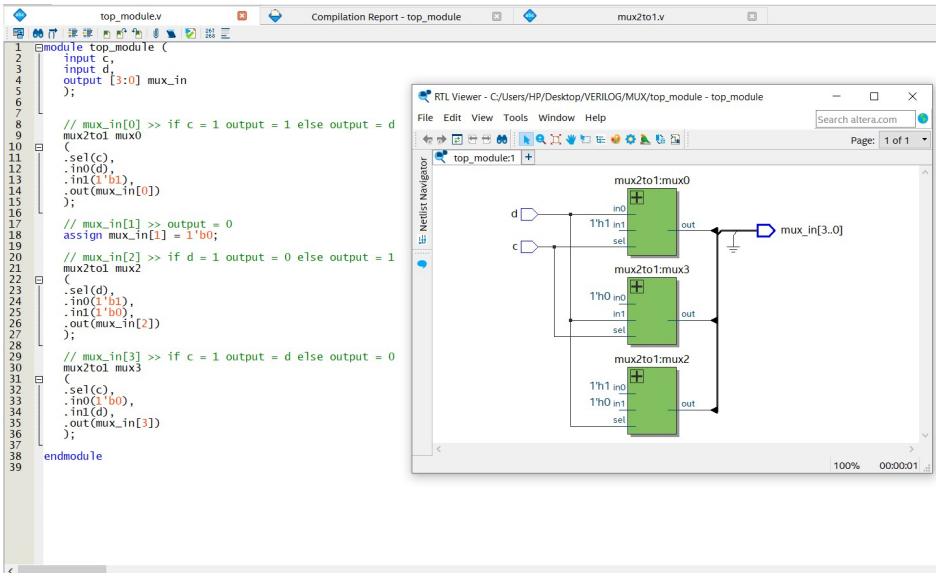

Figure 3: Top Module

1.3 Without using 2-to-1 MUX instances

```
module top_module (
    input a,
    input b,
    input c,
    input d,
    output out
    );

    wire [3:0] mux_in;

    assign mux_in[0] = c ? 1'b1 : d;   // if c output = 1 else output = d
    assign mux_in[1] = 1'b0;   // output = 0
    assign mux_in[2] = d ? 1'b0 : 1'b1;   // if d output = 0 else output = 1
    assign mux_in[3] = c ? d : 1'b0;   // if c output = d else output = 0

    assign out = ({a, b} == 2'b00) ? mux_in[0] :
                 ({a, b} == 2'b01) ? mux_in[1] :
                 ({a, b} == 2'b10) ? mux_in[2] :
                                     mux_in[3];

endmodule
```

Figure 4: Top Module 2

2 Problem B

Design a parameterized N-bit Ripple Carry Adder (RCA) using Verilog.

2.1 Requirements

- Use a parameter WIDTH = 4 to define the number of bits (N).
- Inputs:
  - a: N-bit input operand
  - b: N-bit input operand
  - cin: 1-bit input carry
- Outputs:
  - sum: N-bit sum output
  - cout: Final carry out

2.2 Implementation

`Half Adder >> Full Adder >> N-bit Ripple Carry Adder`

2.2.1 Half Adder

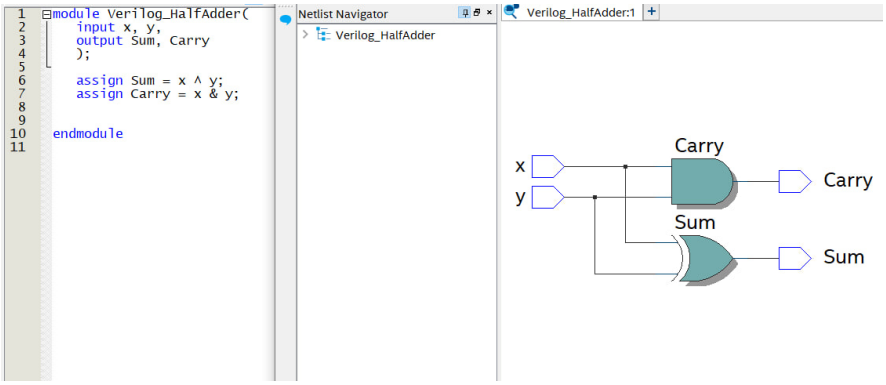

Figure 5: Half Adder

2.2.2 Full Adder

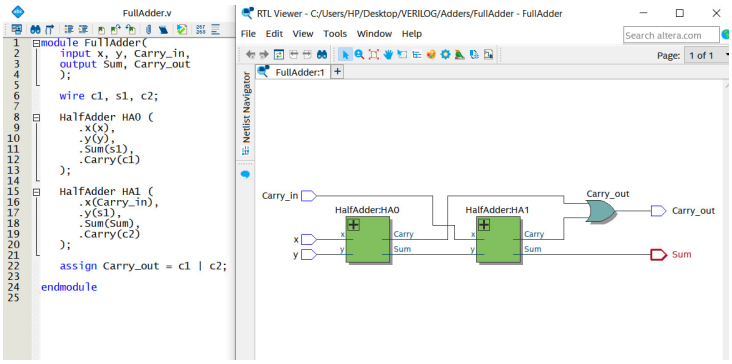

Figure 6: Full Adder

2.2.3 N-bit Ripple Carry Adder

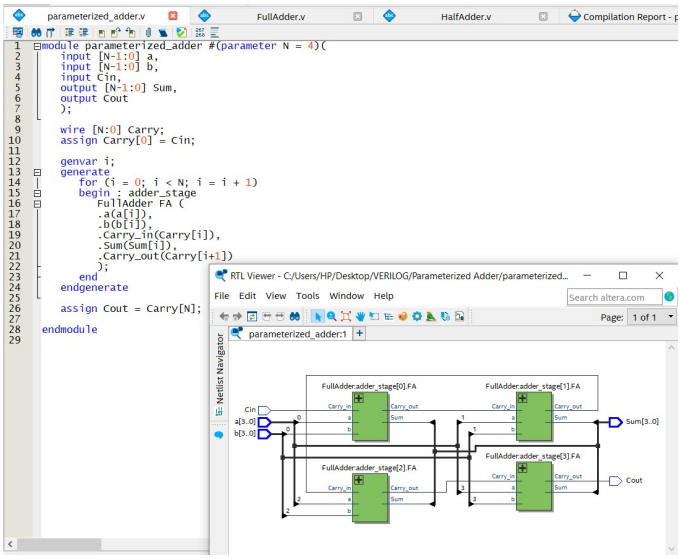

Figure 7: Parameterized Adder

3 Problem C

Create 8 D flip-flops with active high asynchronous reset.
All DFFs should be triggered by the positive edge of clk.

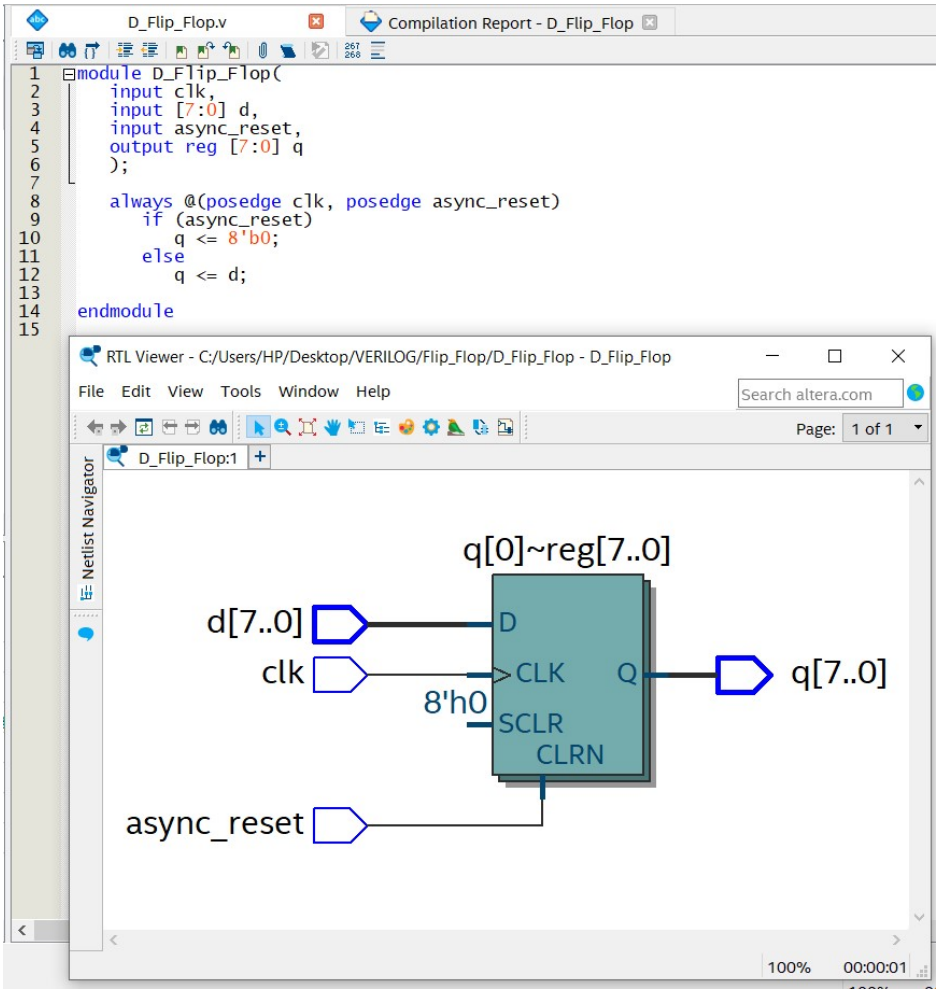

Figure 8: D_Flip_Flop